

Ishayu Ghosh

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EDUCATION

University of California, San Diego

B.S. in Data Science, Minor in Mathematics; GPA: 3.94 – Provost's Honors

La Jolla, CA

Sep 2024 – Jun 2027

EXPERIENCE

de Sa Lab, UCSD

Apr 2025 – Present

ML engineering intern

La Jolla, CA

- Preprocessed raw EEG data across 60 channels using EEGLab: filtering, artifact removal using ICA, epoching around trials, synchronizing with hand motion data
- Designed a two-step classification pipeline: EEGNet (CNN for EEG data) to decode kinematic features from EEG signals, followed by SVM to classify handwritten letters from the decoded kinematics.
- Validated with stratified 5-fold cross-validation, replicating published benchmark results within 1% on a 5-letter subset

UCSD Data Science Student Society

Oct 2024 – Present

Software Developer

La Jolla, CA

- Created responsive design using HTML, Tailwind CSS, implementing animations, which increased society's website traffic by 75%
- Optimized rendering and implemented lazy-loading in React, TypeScript to improve performance, reducing latency by 30%
- Developing backend for society's website using Supabase, Node.js, Express.js, creating points system to improve engagement for 500+ members and event participants.

Infinite Uptime

Jan 2023 – Apr 2023

Data Analyst Intern

Remote

- Processed large-scale industrial data (10,000×400+ records per file) from factory machine monitors using Python (Pandas, NumPy, OS).
- Engineered a pipeline to detect and flag missing velocity features for each monitor, automatically recording missing columns in structured outputs.
- Generalized code into reusable functions across 30+ daily files, reducing manual effort by 80%

PROJECTS

TritonBall Website (UCSD DS3 Project) | *React, TypeScript, CSS*

Apr 2025 – Jun 2025

- Led end-to-end development of TritonBall's website, creating a sleek, interactive platform for UCSD's sports analytics community to access resources, research, and project opportunities
- Built responsive, component-based frontend with React and CSS animations and APIs for calendar integration
- Worked closely with stakeholders through iterative feedback cycles, ensuring requirements were met, resulting in 50% increase in membership

Dubai College Cricket Scoring Web App | *Flask, Pandas, NumPy, Matplotlib, SQL*

Jan 2023 – Sep 2023

- Built a scoring and analytics website for school cricket team
- Used full stack development for live scoring, storing match data, visual analytics
- Used APIs to fetch and store recent world match data, display scoreboard

Cricket Analytics Research Paper | *Young Scientist Journal, Vanderbilt*

Jul 2022 - May 2023

- Analyzed 1,700+ player ratings; found bowling had greater effect on ratings than batting.
- Tested regression, KNN, and decision trees—linear regression performed best (84%)

TECHNICAL SKILLS

Languages: Python, Java, HTML, CSS, JavaScript, TypeScript, SQL, MATLAB

Libraries/Frameworks: Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, Flask, Express.js, React, Node.js, Tailwind, PyTorch, TensorFlow, MNE, SciPy

Tools: AWS, Dataiku, Git, DSA